

**TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION 2011**

**MATHEMATICS
EXTENSION 1**

*Time Allowed – 2 Hours
(Plus 5 minutes Reading Time)*

- All questions may be attempted
- All questions are of equal value
- Department of Education approved calculators and templates are permitted
- In every question, show all necessary working
- Marks may not be awarded for careless or badly arranged work
- No grid paper is to be used unless provided with the examination paper

The answers to all questions are to be returned in separate *stapled* bundles clearly labeled Question 1, Question 2, etc. Each question must show your Candidate Number.

Question 1 (12 Marks)**Marks**

- (a) Find $\frac{d}{dx}(\tan 4x)$. 2
- (b) Find the co-ordinates of the point that divides the interval joining $A(7,2)$ and $B(11,6)$ externally in the ratio 3:5. 2
- (c) Evaluate $\lim_{x \rightarrow 0} \frac{3 \sin x \cos x}{4x}$. 2
- (d) Solve $\cos 2x = -\frac{1}{2}$ for $0 \leq x \leq 2\pi$. 2
- (e) If $x = 1 + \cos \theta$ and $y = 2 - \sin \theta$ find a relationship between x and y only. 2
- (f) Evaluate $\int_0^{2\sqrt{3}} \frac{dx}{4+x^2}$. 2

Question 2 **START A NEW PAGE** (12 Marks)**Marks**

- (a) Using all the letters of the word MATHEMATICS, how many different arrangements can be made. 2
- (b) The temperature, T° centigrade, of a pie t minutes after being placed in an oven is given by the formula $T = 180 + Be^{kt}$. Initially the temperature of the pie is $5^\circ C$ and after 15 minutes the temperature has risen to $40^\circ C$.
- (i) Find the value of the constant B . 1
- (ii) Find the exact value of the constant k . 2
- (iii) Find the temperature of the pie one hour after being placed in the oven. Give your answer correct to the nearest degree. 3
- (c) (i) On the same set of co-ordinate axes draw neat sketches of the graphs $y = x$ and $y = \frac{2}{x-1}$. 2
- (ii) Hence or otherwise solve $x > \frac{2}{x-1}$. 2

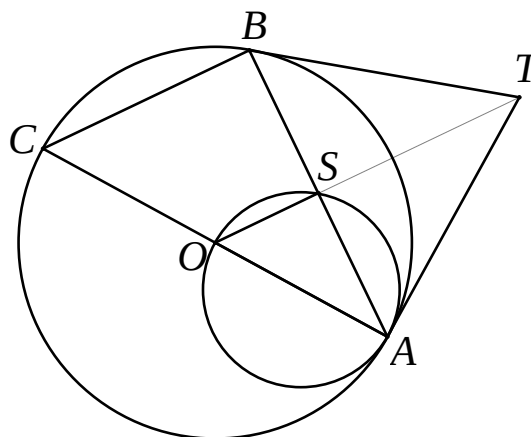
Question 3 **START A NEW PAGE** (12 Marks)**Marks**

- (a) A district squad of 9 netball players is chosen from 3 netball teams (A , B and C). There are 8 players in each of the teams A , B and C .
- (i) If 4 players are chosen at random from team A , 3 from team B and 2 from team C , in how many ways can the district squad be formed? 2
- (ii) Find the probability that Janice from team B and Sarah from team C will be chosen as members of the district squad. 2
- (b) Solve $\sec^2 x + \tan x - 7 = 0$ for $0^\circ \leq x \leq 360^\circ$. Give your answers correct to the nearest minute. 3
- (c) (i) By equating coefficients, find the values of P and Q in the identity 2
 $P(2\sin x + \cos x) + Q(2\cos x - \sin x) \equiv 7\sin x + 11\cos x$.
- (ii) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \frac{7\sin x + 11\cos x}{2\sin x + \cos x} dx$. 3

Question 4 **START A NEW PAGE** (12 Marks)**Marks**

- (a) Evaluate $\int_0^1 \frac{x}{(2x+1)^2} dx$ using the substitution $u = 2x+1$. 4

- (b) Two circles touch at point A . The small circle passes through the centre O of the large circle. AB is a chord of the large circle and cuts the small circle at S . AC is a diameter of the large circle. AT and BT are tangents to the large circle. (See diagram)



- (i) Copy the diagram into your book and prove that CB is parallel to OS . 2
- (ii) Hence prove that $BS = SA$. 2
- (iii) Find the size of $\angle OSA$. 1
- (iv) Prove that the points O , S and T are collinear. 3

Question 5	START A NEW PAGE	(12 Marks)	Marks
(a)	Given that A, B, C and D are the vertices of a cyclic quadrilateral, find the value of $\cos A + \cos B + \cos C + \cos D$.		2
(b)	Use the Principle of Mathematical Induction to prove that $11^n - 2^{2n}$ is divisible by 7 for all integers $n \geq 1$.		4
(c)	The arc of the curve $y = \sin^{-1} x$ that lies in the positive quadrant is rotated one revolution about the y -axis to form the surface of a container.		
(i)	If the container is filled to a depth of h metres, show that the volume, $V \text{ m}^3$, of water in the container is given by: $V = \frac{\pi}{4}(2h - \sin 2h)$.		3
(ii)	The container is being filled at a rate of $6 \text{ m}^3 / \text{hr}$. Calculate the rate at which the depth of water is increasing when the depth is $\frac{\pi}{6} \text{ m}$.		3

Question 6	START A NEW PAGE	(12 Marks)	Marks
(a)	In a small rural community two hobby farms provide eggs for the local grocer. The grocer makes up cartons containing one dozen eggs, always using 8 eggs from farm A and 4 eggs from farm B . Some of the eggs contain two yolks (called a “double-yolker” egg). Eggs from farm A have an 18% probability of being a double-yolker while the probability for farm B is 24%.		
(i)	If an egg is chosen at random from one of the cartons, show that there is a 20% probability that it will be a double-yolker.		2
(ii)	Find the probability that a carton chosen at random will have exactly three double-yolker eggs. Give your answer correct to the nearest percent.		2
(iii)	Find the probability that a carton chosen at random will have at least three double-yolker eggs. Give your answer correct to the nearest percent.		2
(b)	Masses are placed at two points A and B which are 1 metre apart. A 1 kg mass (M) is placed at a point P between A and B . The mass M experiences forces of attraction towards both the points A and B . The force (in Newtons) of the attraction towards A is equal to four times the distance AP while the force of attraction towards point B is equal to the square of the distance PB . Take the origin of the motion at point A and the positive direction of motion in the direction of the ray AB .		
(i)	The mass M at point P is initially x metres from the origin A . Briefly explain why the acceleration, $\ddot{x} \text{ m/s}^2$, of the mass M is given by: $\ddot{x} = x^2 - 6x + 1$.		1
(ii)	If the mass M now starts from rest halfway between A and B , in which direction will it begin to move? Briefly explain your answer.		2
(iii)	Find the speed of the mass M when it first reaches point A .		3

(a) Find the value of the constant term in the expansion of $\left(2y - \frac{1}{y^3}\right)^{20}$.

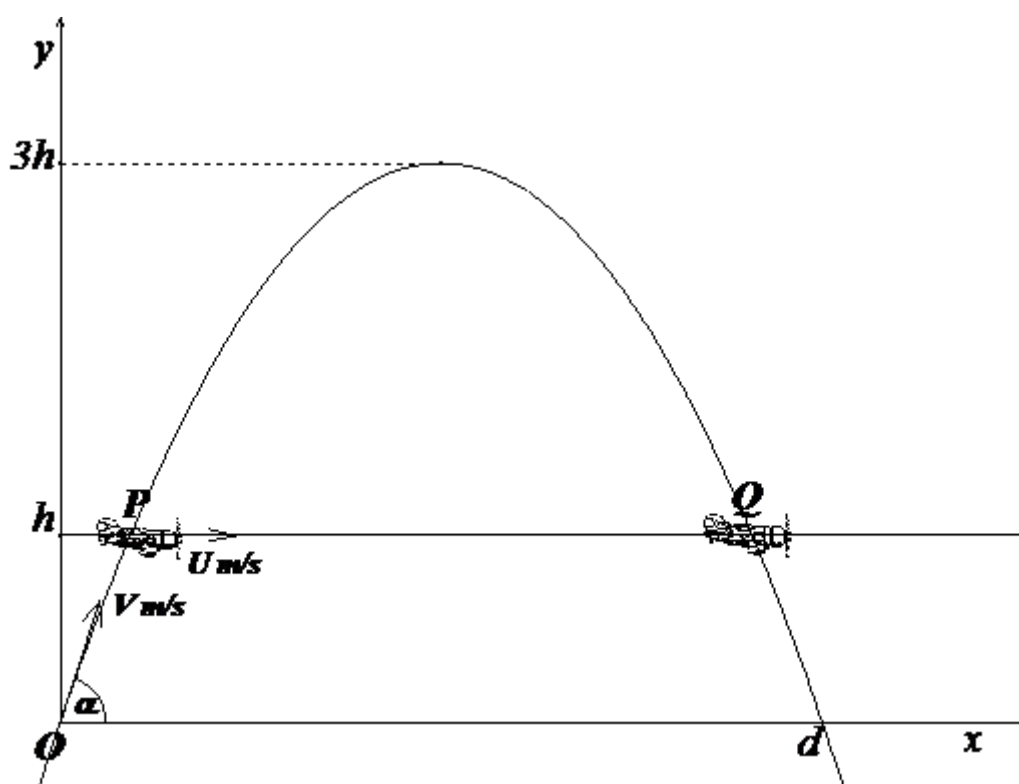
3

- (b) An enemy plane is flying horizontally at height h metres with speed U m/s.

When it is at point P a ground rocket is fired towards it from the origin O with speed V m/s and angle of elevation α .

The rocket misses the plane, passing too late through the point P . However, it goes on to reach a maximum height of $3h$ metres and then on its descent strikes the plane at Q .

With the axes as shown in the diagram, you may assume that the position of the rocket is given by: $x = Vt \cos \alpha$ and $y = -\frac{1}{2}gt^2 + Vt \sin \alpha$, where t is the time in seconds after firing and g is the acceleration due to gravity.



- (i) Show that the initial vertical velocity component ($V \sin \alpha$) of the rocket's speed equals $\sqrt{6gh}$. **2**
- (ii) If the rocket had not struck the plane at Q , it would have returned to the x -axis at a distance d metres from O . **2**
- Show that the horizontal component ($V \cos \alpha$) of the rocket's speed equals $\frac{gd}{2\sqrt{6gh}}$.
- (iii) Show that the equation of the path of the rocket is $y = \frac{12hx}{d} \left(1 - \frac{x}{d}\right)$. **2**
- (iv) If the horizontal component of the rocket's speed is $100(3 + \sqrt{6})$ m/s, find the time taken by the rocket to strike the plane at Q , in terms of d . **2**
- (v) Find the speed of the enemy plane. **1**

30 TRIAL
MATHEMATICS Extension 1 : Question...1...

2011

Suggested Solutions

Marks

Marker's Comments

1) $\frac{d(\tan 4x)}{dx} = 4 \sec^2 4x$

2

if they integrated
or use inverse trig
→ 0 marks

2) A(-7, 2) B(1, 6)

$3^2 - 5$

$P = \left(\frac{7x-5+3 \times 11}{3x-5}, \frac{2x-5+3 \times 6}{3x-5} \right)$

1/2

* If they did an
internal division and
got $(8\frac{1}{2}, 3\frac{1}{2}) \rightarrow 1 \text{mk}$

$= \left(\frac{-35+33}{-2}, \frac{-10+18}{-2} \right)$

1/2

$= \left(\frac{2}{-2}, \frac{8}{-2} \right)$

1/2

$= (1, -4)$

1/2

3) $\lim_{x \rightarrow 0} \frac{3 \sin x \cos x}{4x}$

1/2

$= \lim_{x \rightarrow 0} \frac{6 \sin x \cos x}{8x}$

1/2

$= \lim_{x \rightarrow 0} \frac{3 \sin 2x}{8x}$

1/2

$= \lim_{x \rightarrow 0} \frac{\sin(2x) \cdot 3}{2x} \cdot \frac{2x}{x} \cdot \frac{1}{4}$

$= 1 \times \frac{3}{4}$

1/2

$= \frac{3}{4}$

or $\lim_{x \rightarrow 0} 3 \times \frac{\sin x}{x} \times \frac{\cos x}{1} \times \frac{1}{4}$

1/2

$= 3 \times 1 \times 1 \times \frac{1}{4}$

1/2

$= \frac{3}{4}$

4) $\cos 2x = -\frac{1}{2}$ for $0 \leq x \leq 2\pi$

1/2

* If they used
general solutions, then
they had to use
 $\cos^{-1}(-\frac{1}{2}) = \frac{2\pi}{3}$ as
 $\cos^{-1} x$ is defined
 $0 \leq x \leq 2\pi$

$2x = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}, 3\pi - \frac{\pi}{3}, 3\pi + \frac{\pi}{3}$

$= \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3}$

$\therefore x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$

1/2

each

2/2

MATHEMATICS Extension 1 : Question...1...

2011

Suggested Solutions

Marks

Marker's Comments

5) $x = 1 + \cos \theta$ $y = 2 - \sin \theta$

1/2

$x - 1 = \cos \theta$ $2 - y = \sin \theta$

1/2

$\cos^2 \theta + \sin^2 \theta = 1$

1

$\therefore (x-1)^2 + (2-y)^2 = 1$

or $2 - \sqrt{2x-x^2} = y$ or $x^2 - 2x + y^2 - 4y = 1$

1/2

6) $\int_0^{2\sqrt{3}} \frac{dx}{4+x^2} = \frac{1}{2} \left[\tan^{-1} \left(\frac{x}{2} \right) \right]_0^{2\sqrt{3}}$

1/2

$= \frac{1}{2} \left(\tan^{-1} \frac{2\sqrt{3}}{2} - \tan^{-1} 0 \right)$

1/2

$= \frac{1}{2} \left(\frac{\pi}{3} - 0 \right)$

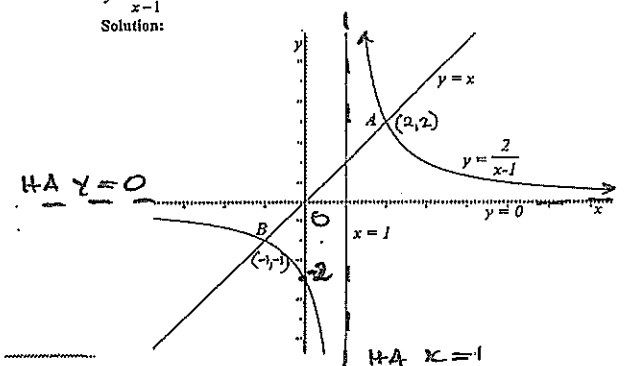
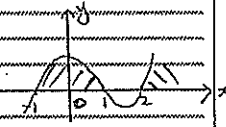
1/2

$= \frac{\pi}{6}$

2011 TRIAL HSC MATHEMATICS Extension 1 : Question 2

Suggested Solutions	Marks	Marker's Comments
Q2 a) MATHEMATICS No. of arrangements = $\frac{11!}{2! \times 2! \times 2!}$ $= 4989600$	1	
b) (i) $T = 180 + Be^{kt}$ When $t=0$, $T=5$ $5 = 180 + Be^0$ $B = -175$	1	
(ii) $T = 180 - 175e^{kt}$ When $t=15$, $T=40$ $40 = 180 - 175e^{15k}$ $175e^{15k} = 140$ $e^{15k} = \frac{140}{175}$ $e^{15k} = \frac{4}{5}$ $15k = \ln\left(\frac{4}{5}\right)$ $k = \frac{1}{15} \ln\left(\frac{4}{5}\right)$	1	$-\frac{1}{15} \ln\left(\frac{4}{5}\right)$ lost $\frac{1}{2}$ mark
(iii) $T = 180 - 175e^{\frac{1}{15} \ln\left(\frac{4}{5}\right)t}$ When $t=60$ $T = 180 - 175e^{\frac{1}{15} \ln\left(\frac{4}{5}\right) \times 60}$ $= 180 - 175e^{4 \ln\left(\frac{4}{5}\right)}$ $= 108.32$ $\therefore \text{Temp.} = 108^\circ\text{C}$ (nearest degree)	1	No Celsius lost $\frac{1}{2}$ mark

page 2 MATHEMATICS Extension 1 : Question 2

Suggested Solutions	Marks	Marker's Comments
Q2 (c) (i) On the same set of co-ordinate axes draw neat sketches of the graphs $y=x$ and $y=\frac{2}{x-1}$. Solution: 	2	$\frac{1}{2}$ mark for $y=\frac{2}{x-1}$ with y intercept -2 $\frac{1}{2}$ mark for $y=x$ $\frac{1}{2}$ mark for H.A. $y=0$ $\frac{1}{2}$ mark for V.A. $x=1$
(c) (ii) Solve $x > \frac{2}{x-1}$ At A and B $x=2$ $x^2 - x - 2 = 0$ $(x-2)(x+1) = 0$ $x = -1$ or 2 $\therefore$ For $x > \frac{2}{x-1}$ $-1 < x < 1$ or $x > 2$ OR $x > \frac{2}{x-1}$ $\frac{2}{x-1} < x$ $\frac{2}{x-1} \times (x-1)^2 < x(x-1)^2$ $x(x-1)^2 - 2(x-1) > 0$ $(x-1)[x(x-1)-2] > 0$ $(x-1)(x+1)(x-2) > 0$ $\therefore -1 < x < 1$ or $x > 2$ OR $x - \frac{2}{x-1} > 0 \Rightarrow \frac{x^2 - x - 2}{x-1} > 0$ $\Rightarrow (x-1)(x-2)(x+1) > 0$ 	2	1 mark for each region

EXTENSION MATHEMATICS: Question 3

Suggested Solutions

Marks

Marker's Comments

a) (i) Number of ways

$$= {}^8C_4 \times {}^8C_3 \times {}^8C_2 = 109\,760$$

(ii) P(sarah and Janice)

$$= \frac{{}^8C_4 \times {}^7C_2 \times {}^2C_1}{{}^8C_4 \times {}^8C_3 \times {}^8C_2} = \frac{10290}{109\,760} = \frac{3}{32}$$

OR

$$P(S \text{ and } J) = P(S) \times P(J)$$

$$= \frac{3}{8} \times \frac{2}{8}$$

$$= \frac{3}{32}$$

b) $\sec^2 x + \tan x - 7 = 0$

$$\tan^2 x + \tan x - 6 = 0$$

$$(\tan x + 3)(\tan x - 2) = 0$$

$$\tan x = -3 \text{ or } \tan x = 2$$

Reference angles: $71^\circ 34'$ and $63^\circ 26'$

Hence Solution Set is:

$$\{108^\circ 26'; 288^\circ 26'; 63^\circ 26'; 243^\circ 26'\}$$

General Solution: $x = n\pi + \tan^{-1}(2)$
or $x = n\pi + \tan^{-1}(-3)$

For $[0, 360^\circ]$, start with $n=0, 1, 2$ etc

$$1 + \sin x \cos x - 7 \cos^2 x = 0$$

$$\sin^2 x + \sin x \cos x - 6 \cos^2 x = 0$$

$$\sin^2 x + \sin x \cos x - 6 \cos^2 x = 0$$

EXTENSION MATHEMATICS: Question 3

Suggested Solutions

Marks

Marker's Comments

(c) (i) Expanding and factoring:

$$(2P-Q)\sin x + (P+2Q)\cos x = 7\sin x + 11\cos x$$

Equating coefficients of like terms:

$$2P-Q = 7 \quad \dots (i)$$

$$P+2Q = 11 \quad \dots (ii)$$

$$(ii) \times 2: 2P+4Q = 22 \quad \dots (iii)$$

$$(iii) - (i): 5Q = 15$$

$$Q = 3$$

$$\Rightarrow P = 5$$

(ii) From (i):

$$\int_0^{\pi/2} \frac{7\sin x + 11\cos x}{2\sin x + \cos x} dx = \int_0^{\pi/2} \frac{5(2\sin x + \cos x) + 3(2\cos x - \sin x)}{(2\sin x + \cos x)} dx$$

$$= \int_0^{\pi/2} \left[5 + \frac{3(2\cos x - \sin x)}{(2\sin x + \cos x)} \right] dx$$

$$= \int_0^{\pi/2} \left[5 + 3 \left\{ \frac{d}{dx} (2\sin x + \cos x) \right\} \right] dx$$

$$= \left[5x + 3 \ln |2\sin x + \cos x| \right]_0^{\pi/2}$$

$$= \frac{5\pi}{2} + 3 \ln |2 \times 1 + 0| - [5 \times 0 + \ln |2 \times 0 + 1|]$$

$$= \frac{5\pi}{2} + 3 \ln 2$$

• Every mistake $-\frac{1}{2}$
• Careless mistakes very evident
• Use of substitution technique popular!
• Using values from (i)
• for splitting fraction and simplifying
• for recognising $\frac{f'(x)}{f(x)}$ formula
• for each integral
• for final answer
• $\frac{5\pi}{2} + 3 \ln 2 \dots \text{max } 2\frac{1}{2}$
• generally well answered

Suggested Solutions

Marks

Marker's Comments

$$\frac{1}{2} \int \frac{2x dx}{(2x+1)^2}$$

$$u = 2x+1 \quad du = 2dx$$

$$x=0, u=1; x=1, u=3$$

$$= +\frac{1}{2} \int \frac{u-1}{u^2} \frac{du}{2}$$

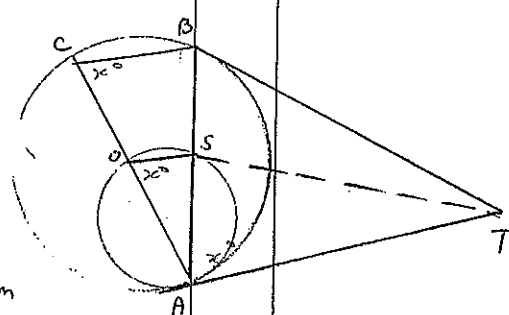
$$= \frac{1}{4} \int \frac{u-1}{u^2} du$$

$$= \frac{1}{8} \int \frac{2u du}{u^2} - \frac{1}{4} \int \frac{du}{u^2}$$

$$= \frac{1}{8} \left[\ln u^2 \right] + \left[\frac{1}{4u} \right]$$

$$= \frac{1}{8} \ln 3^2 + \frac{1}{4} \left(\frac{1}{3} - 1 \right)$$

$$= \frac{1}{4} \ln 3 - \frac{1}{6}$$



Method 1 A.K.A
i) $\angle CBA = \angle TAB$
(angle between tangent & chord equals to
angle at circumference in alternate segment)

Similarly $\angle TAB = \angle OSA$
A.O.S

$\therefore \angle CBA = \angle OSA$

$\therefore CB \parallel OS$ (2 lines are parallel if
their corresponding angles
are equal)

Method 2

When 2 circles touch at a point, line
through centre of one circle to point of contact
will pass through centre of second circle

Students can't
prove $\angle CBA = \angle OSA$
correctly
can't get the
last mark.

Students can't
prove A.O. is
diameter of
small circle
max 1m only

Suggested Solutions

Marks

Marker's Comments

$\therefore AO$ is diameter of small circle

$\angle OSA = 90^\circ$ (angle in semi-circle)

Since CO is diameter of big circle
similarly $\angle CBA = 90^\circ$

$\therefore \angle OSA = \angle CBA$

$\therefore CB \parallel OS$ (2 lines are parallel
if corresponding angles are equal)

ii) Method 1 $CO = AO$ (radii of same circle)

$\therefore CO/AO = 1$

$CB \parallel OS$ (proved in i)

$\therefore \frac{AS}{BS} = \frac{CO}{AO} = 1$ (line parallel to one side
of triangle divides the other
2 sides in same ratio)

$\therefore AS = BS$

Method 2

In $\triangle OAS$, $\triangle CBA$

$\angle OSA = \angle CBA = 90^\circ$ (proved in i)

$\angle OAS = \angle CAB$

$\therefore \triangle OAS \parallel \triangle CBA$ (equiangular)

$\frac{AO}{AC} = \frac{AS}{BS}$ (corresponding sides of similar
triangles are in same ratio)

$\frac{AO}{AC} = \frac{1}{2}$ (radius is half diameter
in big circle)

$\therefore \frac{AS}{BS} = \frac{1}{2}$

$\frac{AS}{AS+BS} = \frac{1}{2}$

$\therefore AS = BS$

many students
forgot same ratio.
- 1/2 m

Suggested Solutions

Marks

Marker's Comments

Method 3

$$\angle OSA = 90^\circ \quad (\text{proved in i})$$

$$OS \perp AB$$

$\therefore BS = AS$ (line from centre of circle perpendicular to chord bisects it)

$$\text{ii) } \angle CBA = 90^\circ \quad (\text{angle in semi-circle})$$

$$\angle OSA = \angle CBA \quad (CB \parallel OS, \text{ corresponding angles equal})$$

$$\therefore \angle OSA = 90^\circ$$

$$\text{v) } \angle \triangle BTS \text{ r } \triangle ATS$$

$TA = TB$ (tangents to a circle from an external point are equal)

TS is common

$$BS = AS \quad (\text{proved in ii})$$

$$\therefore \triangle BTS \cong \triangle ATS \quad (SSS)$$

$$\therefore \angle TSB = \angle TSA \quad (\text{corresponding angles of congruent triangles})$$

$$\angle BSA = 180^\circ \quad (\text{angle sum of straight angle})$$

$$\therefore \angle TSB = \angle TSA = \frac{180^\circ}{2} = 90^\circ$$

$$\angle OSA = 90^\circ \quad (\text{proved in ii})$$

$$\angle OSA + \angle TSA = 90^\circ + 90^\circ = 180^\circ$$

$\angle OST$ is a straight angle

$\therefore O, S, T$ are collinear $\#$.

1m

1m

1m

1m

$\frac{1}{2}$ m

$\frac{1}{2}$ m

1m

MANY has proved in part i or ii

Some students prove $\angle TAS = \angle TBS$ instead of TS
 $\triangle TSB \cong \triangle TSA$ (SAS)

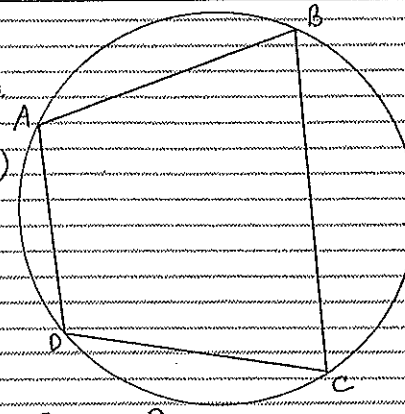
Some students assumed OST is st.
 $\angle BST = \angle OSA$
(vertically opp. angles)

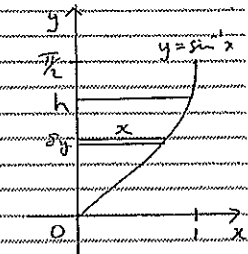
not $\frac{1}{2}$ m

or

$\triangle OAT$

must state straight angle for collinear
 $-\frac{1}{2}$ m

11 TRIAL JRAHS MATHEMATICS Extension 1 : Question 5...		
Suggested Solutions	Marks	Marker's Comments
 $C = 180^\circ$ (Opposite angles of a cyclic quadrilateral are supplementary) $A = \cos(180^\circ - C)$ $A = -\cos C$ $B = -\cos D$ $\therefore A + \cos B + \cos C + \cos D$ $= -\cos C + \cos D + \cos C + \cos D = 0$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	$\frac{1}{2}$ for geometry reason
To prove $11^n - 2^{2n}$ is divisible by 7 for $n \geq 1$ ep.1 Consider $n=1$ LHS = $11^1 - 2^2$ $= 11 - 4$ $= 7$ which is divisible by 7 True for $n=1$ ep.2 Assume true for $n=k$ where $k \in \mathbb{N}$ i.e. $11^k - 2^{2k} = 7A$ where A integer ITP $11^{k+1} - 2^{2(k+1)} = 7B$ some other integer B Now $11^{k+1} - 2^{2(k+1)} = 11 \cdot 11^k - 4 \cdot 2^{2k}$ $= 11(7A + 2^{2k}) - 4 \cdot 2^{2k}$ by Assumption $= 11 \cdot 7A + 2^{2k}(11 - 4)$ $= 7(11A + 2^{2k})$ $= 7B$ where $B = 11A + 2^{2k}$ is an integer	$\frac{1}{2}$ $\frac{1}{2}$ 2	Although most people had the general idea, there was some very slack presentation

2011 TRIAL JRAHS MATHEMATICS Extension 1 : Question 5... (cont)		
Suggested Solutions	Marks	Marker's Comments
Thus, if true for $n=k$, also true for $n=k+1$ Step 3 Using steps 1 and 2, by the Principle of Mathematical Induction, thus proved <u>$11^n - 2^{2n}$ is divisible by 7 for $n \geq 1$</u>	$\frac{1}{2}$	
c) i) $\delta V = \pi x^2 \delta y$ $V = \pi \int_0^h x^2 dy$ $= \pi \int_0^h \sin^2 y dy$ $= \frac{\pi}{2} \int_0^h (1 - \cos 2y) dy$ $= \frac{\pi}{2} \left[y - \frac{1}{2} \sin 2y \right]_0^h$ $= \frac{\pi}{2} \left(h - \frac{1}{2} \sin 2h - 0 + 0 \right)$ $= \frac{\pi}{4} (2h - \sin 2h)$  ii) Find $\frac{dh}{dt}$ given that $\frac{dV}{dt} = 6 \text{ (m}^3/\text{hr)}$ $\frac{dh}{dt} = \frac{dV}{dt} \times \frac{dh}{dV} = \frac{dV}{dt} \cdot \frac{dV}{dh}$ (Chain Rule) $\frac{dV}{dh} = \frac{\pi}{4} (2 - 2 \cos 2h) = \frac{\pi}{2} (1 - \cos 2h)$ $= \frac{\pi}{2} (1 - \frac{1}{2})$ when $h = \pi/6$ $= \pi/4$ $\therefore \frac{dh}{dt} = \frac{6/(\pi/4)}{\pi/4} = \frac{24}{\pi}$ $\therefore$ Depth increasing at rate of $\frac{24}{\pi} \text{ m/hr}$	1 1 1 1	$\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$ It would have been nice to see "Chain Rule" written Too few people did not finish the answer in sentence form

MATHEMATICS Extension 1: Question 6

Suggested Solutions

Marks

Marker's Comments

6(a) (i) Farm A: uses 8YYs, Farm B uses 4YYs

$$P(E=YY) = P(AYY \text{ or } BYY)$$

$$= P(AYY) + P(BYY)$$

$$= \frac{8}{100} \times \frac{18}{100} + \frac{4}{100} \times \frac{24}{100}$$

$$= \frac{2}{25} \times \frac{9}{50} + \frac{1}{25} \times \frac{6}{25}$$

$$= \frac{3}{25} + \frac{2}{25} = \frac{5}{25}$$

$$= \frac{1}{5} = 0.2$$

∴ Probability of YY is 20% qsd.

i) Now $P(YY) = p = 0.2$
 $\therefore P(\overline{YY}) = q = 0.8$
 Using $(q+p)^n = (0.8+0.2)^{12}$ Binomial Prob

$$P(X=3YYs) = \binom{12}{3} q^9 p^3 = \binom{12}{3} (0.8)^9 (0.2)^3$$

$$= 0.23622320 \dots$$

$$\approx 24\% \text{ (nearest \%)}$$

$$P(X \geq 3YYs) = 1 - [P(X=0) + P(X=1) + P(X=2)]$$

$$= 1 - \left[\binom{12}{0} (0.8)^{12} + \binom{12}{1} (0.8)^{11} (0.2) + \binom{12}{2} (0.8)^{10} (0.2)^2 \right]$$

$$= 1 - [0.068719 \dots + 0.206158 \dots + 0.283467 \dots]$$

$$= 1 - 0.558345 \dots$$

$$= 0.441654 \dots$$

$$P(X \geq 3YY) \approx 44\% \text{ (nearest \%)}$$

1/2

Many misinterpreted the Q, and got "24%"

1/2 For (12/3)

1/2

1/2

2

2

2

MATHEMATICS Extension 1: Question 6

Suggested Solutions

Marks

Marker's Comments

6(b) A — P — B
 0 x 1-x
 Particle

(i) mass: $m=1$
 Resultant force $= 1x = F_B - F_A$
 towards B $1x = (1-x)^2 - 4x^2$
 $\therefore x^2 = 1-2x+x^2-4x^2$
 $1x^2 = x^2 - 6x + 1$ qsd.

$t=0 \quad x=\frac{1}{2} \quad v=0$
 (ii) $\therefore x = \left(\frac{1}{2}\right)^2 - 6\left(\frac{1}{2}\right) + 1 = \frac{1}{4} - 3 + 1$
 $x = -\frac{11}{4} < 0$
 $\therefore$ acceleration is $-1\frac{3}{4} \text{ m/s}^2$
 $\therefore$ Applied force is to the left ($x < 0$) but as $v=0$
 $\therefore$ motion of particle is towards the left

(iii) $x = \frac{1}{2}v^2 = x^2 - 6x + 1 = (x-3)^2 - 8$
 $\therefore \frac{1}{2}v^2 = \frac{1}{2}x^2 - 3x + 1$
 $x = \frac{1}{2} \quad v=0 \Rightarrow C = 5$
 $\therefore v^2 = \frac{1}{2}x^2 - 3x + 5$
 $\therefore v = \frac{1}{\sqrt{2}} \sqrt{x^2 - 6x + 10}$
 $\therefore v = -\sqrt{\frac{5}{2}}$ for $0 \leq t \leq 5$ ($x=0$)

$\therefore$ the speed is $\sqrt{\frac{5}{2}} \text{ m/s}^{-1}$ (0.645...)
 OR $\int d\left(\frac{1}{2}v^2\right) = \int (x^2 - 6x + 1) dx$
 $\frac{1}{2}v^2 = \frac{1}{3}x^3 - 3x^2 + x$
 $= 0 - \left(\frac{1}{24} - \frac{3}{4} + 1\right)$
 $= \frac{5}{24}$
 $v^2 = \frac{5}{12}$ etc

1/2

1/2

1

1/2

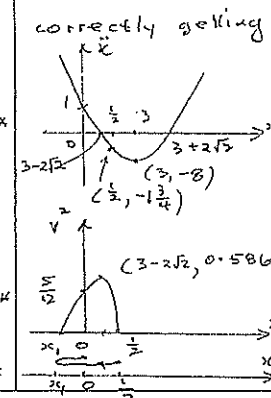
1/2

-1/2 if no mention of mass of 1kg

1/2 For $-1\frac{3}{4}$

starting from rest (towards A)
 1/2 For Force to the L
 1/2 For initially at rest
 1/2 For motion to A.

$F_A = 2 \text{ N}$ $F_B = \frac{1}{2} \text{ N}$
 $F_A > F_B$
 $\therefore$ resultant Force applied to the L



correctly getting to

Question (1)

[3]

(a)

$$(2y - y^{-3})^{20}$$

$$T_{k+1} = \binom{20}{k} (20y)^{20-k} (-y^{-3})^k \quad (1)$$

$$= (-1)^k 2^{20-k} \binom{20}{k} y^{20-4k} \quad (1)$$

For constant term

$$20 - 4k = 0 \Rightarrow k = 5$$

$\therefore T_6$ is the constant term

$$\begin{cases} T_6 = -2^{15} \binom{20}{5} \\ = -15504 \times 2^{15} \\ = -508035072 \end{cases} \quad (1) \quad [2]$$

(b) $y = (v \sin \alpha)t - \frac{gt^2}{2} \quad (3)$

(i) $y = v \sin \alpha - gt \quad (1)$

For max. height $\dot{y} = 0$

$$\therefore t = \frac{v \sin \alpha}{g} \quad (2) \quad (1)$$

Substitute (2) into (3)

We have.

$$y_{\max} = \frac{v^2 \sin^2 \alpha}{2g}, \text{ but } y_{\max} = 3R$$

$$\therefore v^2 \sin^2 \alpha = 6gh.$$

$$v \sin \alpha = \sqrt{6gh}. \quad (1)$$

[A]

(ii) Range = d

[2]

When $y = 0$

$$\text{i.e. } t(v \sin \alpha - \frac{gt}{2}) = 0$$

$$\therefore T(\text{time of flight}) = \frac{2v \sin \alpha}{g} \quad (1)$$

$$R = (v \cos \alpha) \left(\frac{2v \sin \alpha}{g} \right)$$

$$\text{but } R(\text{range}) = d$$

$$\therefore d = v \cos \alpha \left(\frac{2}{g} \sqrt{6gh} \right)$$

$$\therefore v \cos \alpha = \frac{gd}{2\sqrt{6gh}} \quad (1)$$

(iii)

[2]

$$x = v \cos \alpha t$$

$$y = (v \sin \alpha)t - \frac{gt^2}{2}$$

Eliminate t

We have

$$y = x \left(\frac{\sin \alpha}{\cos \alpha} \right) - \frac{gx^2}{2} \frac{1}{(v \cos \alpha)^2}$$

$$\downarrow \frac{v \sin \alpha}{v \cos \alpha} = \frac{12Rx}{d}$$

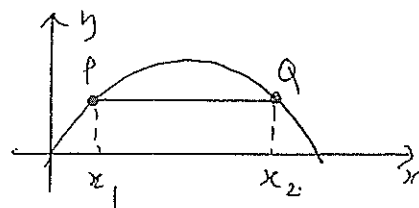
$$\therefore y = \frac{12Rx}{d} - \frac{gx^2}{2} \times \frac{24gh}{g^2 d^2}$$

$$\therefore y = \frac{12Rx}{d} \left(1 - \frac{x}{d} \right) \quad (1)$$

(iv) From (iii) when $y = h$

$$h = \frac{12h}{d} x \left(1 - \frac{x}{d}\right)$$

$$\therefore 12x^2 - 12dx + d^2 = 0 \quad \text{--- (1)}$$



Let the roots be x_1, x_2 .

$$x_1 + x_2 = \frac{12d}{12} = d$$

$$x_1 x_2 = \frac{d^2}{12}$$

$$\begin{aligned} \therefore (x_2 - x_1)^2 &= (x_2 + x_1)^2 - 4x_1 x_2 \\ &= d^2 - \frac{4}{12} d^2 \\ &= \frac{2d^2}{3} \end{aligned}$$

$$\text{i.e. } PQ = (x_2 - x_1) = \frac{\sqrt{6}d}{3} \quad \text{--- (1)}$$

or Use quad. formula

$$x = \frac{12d \pm \sqrt{144d^2 - 48d^2}}{24}$$

$$\text{where } x_2 - x_1 = \left[\left(\frac{3+\sqrt{6}}{6} \right) - \left(\frac{3-\sqrt{6}}{6} \right) \right] d$$

$$x = 100(3+\sqrt{6})t$$

$$\therefore \left(\frac{3+\sqrt{6}}{6} \right) d = x_Q$$

$$\therefore \left(\frac{3+\sqrt{6}}{6} \right) d = 100(3+\sqrt{6})t \quad \text{--- (1)}$$

$$\Rightarrow t = \frac{d}{600}$$

(v) [1]

Distance = Speed $\times$ time

$$\frac{\sqrt{6}d}{3} = u \times \frac{d}{600}$$

$$\therefore \frac{\sqrt{6}}{3} = \frac{u}{600}$$

$$u = 200\sqrt{6} \text{ (m/s)}$$

$$\therefore \text{speed} = 490 \text{ m/s.}$$